

AI Operations (AIOps) Q1

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1 Technical Debt Diagnosis

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- a) This part can be explained by **Entanglement (CACE)** as changing the rounding logic of the estimated delivery time unexpectedly affected the favorite restaurants recommendations, showing that changes in one part can affect another part of the system.
- b) This part can be explained by **Undeclared Consumers** as the marketing dashboard team was silently using the model's raw output without the ML team's knowledge, creating an unknown dependency on the model.
- c) This part can be explained by **Configuration & Glue-Code Debt** as the training pipeline consisted of 14 undocumented shell scripts with no orchestration tool, making the pipeline difficult to manage hence creating tangled code.

2. Mitigation for (a):

Use **MLflow** to track the experiments, including the parameters and metrics used by the model. This makes changes to the model and its configuration easier to track and compare, reducing the risk of unexpected effects from undocumented changes.